Algebra 1: 2nd Semester Final Review
Chapter 8 & 9

State the transformations that occurred from $p(x) = x^2$ to the given function.

1. $g(x) = -2(x-4)^2 + 5$

2. $f(x) = (x+3)^2 - 2$

Find the zeros of the function.

3. $f(x) = 3x^2 - 6x - 45$

4. $k(x) = 2(x-1)(x+7)$

5. $g(x) = x^3 - 16x$

Solve the equation:

6. $r(r-10) = 0$

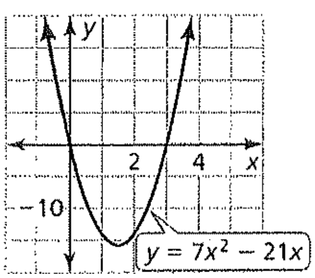
7. $(h-8)^2 = 0$

8. $-2v(v+1) = 0$

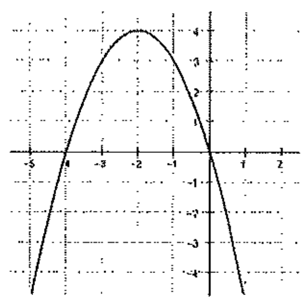
9. $(x-2)(3x-1) = 0$

10. Find the solutions of each function.

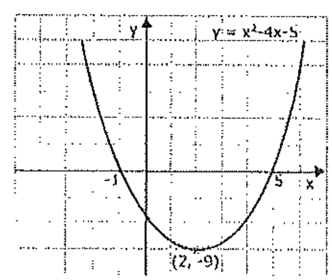
a)



b)



c)

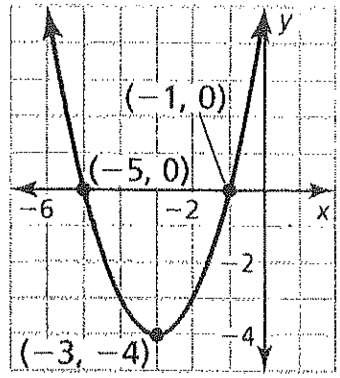


Find the vertex and axis of symmetry and vertex of the graph of the function.

12. $f(x) = 2x^2 - 20x + 51$

13. $f(x) = -\frac{1}{2}(x-3)^2$

11. Consider the graph of the function f. Find the following:



Domain:

Range:

Zeros:

Vertex:

AS:

Write the function in vertex form:

Solve the equation using any method. Leave answer in simplified radical form.

16. $(2x + 1)^2 = 36$

17. $(4x + 5)^2 = 9$

18. $x^2 - 3x - 40 = 0$

19. $-5x^2 = -25$

How many real solutions does each quadratic function have? (Hint: find the discriminate)

20. a. $4x^2 + 4x + 1$

b. $2x^2 - x - 1 = 0$

21. Find the vertex: $k(x) = -(x-3)(x+2)$

22. A ball is dropped out a window. The function $h = -16t^2 - 8t + 25$ represents the height (in feet) of the ball after t seconds. What is the initial height of the ball?